Homework 1 & 2:

Capillarity: Measuring Surface Tension, Minimal Surfaces and Laplace Pressure

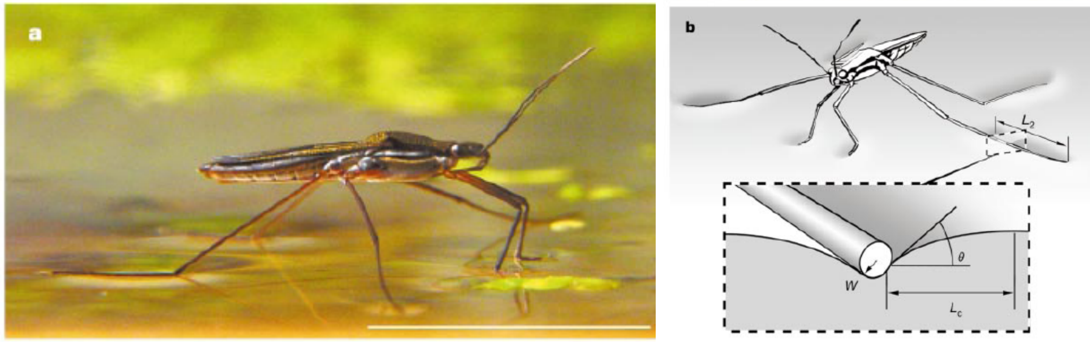
(**Original problem**(s)**: do not repost anywhere without permission**)

(**Problem 1-3 HW 1, & problem 3 optional for UGs; Problem 4-6, HW 2, & prob 6 optional for UGs**).

(1) **Watch these three videos.** (**a**) https://youtu.be/MUlmkSnrAzM?si=Vjz7-Z9QHXcHST-N (**b**) https://youtu.be/XDQ21ngtpkI?si=T6HFs5dvztL1bKAe (**c**) https://youtu.be/QiZ--8tkMdo?si=bozLI6js3rMl4h73 Two questions: (i) Among the phenomena discussed in video (a), pick any two and run the experiment in your kitchen. Feel free to include links to videos you make, but more importantly, summarize the idea, the experiment, the observation, and how well your results compared to what was shown in the video. (ii) What is your favorite dimensionless number? (iii) list as many characteristic lengths as you can from the video, especially those with surface tension involved in the definition.

(2) **Butterfly Proboscis:** Butterflies (and many other insects) sip nectar (blood, sap, water, *etc.*) using their proboscis, a straw-like capillary tube, and though they don't probably know this, interfacial phenomena play a crucial role in their survival. For argument's sake, we can estimate the radius of a long cylindrical capillary needed for bringing water to a height of 1 mm, 10 mm, and 100 mm without the need for external pumping. State any assumptions you make to calculate the estimated capillary radius. What is the hydrostatic pressure at 0.1 mm, 5 mm, and 50 mm under water? How does it compare to atmospheric pressure? Colloidal and Interfacial Phenomena in Proboscis could be a nice topic for a final project. For a more detailed discussion on microfluidics involved in proboscis, see the following reference:
http://jeb.biologists.org/content/217/12/2130?ijkey=5fc423a6dd6efe3313074666b54a85b53a1a603e&keytype2=tf_ipsecsha

(3) **Water Striders:** (a) An insect can walk on water as long as an additional vertical force counterbalances its weight W. In class, we discussed how floating objects deform the surface of a fluid, such as water, how to calculate the curvature of such a surface, and the Laplace pressure associated with curved interfaces. Using this knowledge, discuss how the maximum weight that can be supported depends upon the surface tension of the underlying fluid. What would be a good upper limit to the weight that can be supported by a cylindrical rod placed on water? [Water Strider & relevant Interfacial Phenomena could be another good theme for a final project].
(b) Water striders have thin hydrophobic legs, and penetrating the water surface requires a force equal to the surface tension of water times the perimeter of the leg. Carbon Nanotube forests, on the other hand, have multiple cylindrical rods that can together form a superhydrophobic surface. In general, for elastic rods, buckling occurs when the compressive force exceeds the Euler's critical load $F_{Euler} = (\pi/2)^2 ER^4/L^2$ where E is the elastic modulus, R is the radius of the rod, and L is the length of the rod. What is the critical length of the rod beyond which buckling occurs under capillary forces? Compare this critical length to the capillary length of the fluid. Estimate the critical length for human hair (E = 4 GPa, R = 40 micron), assuming the hair is brought in contact with water.



(4) **"The Cheerios Effect":** http://people.maths.ox.ac.uk/vella/Vella2005.pdf Attraction between two plates dipped into a fluid depends upon interfacial deformation (and in turn on surface tension, contact angle, *etc.*) Re-derive the expressions (1)-(6) given in the paper titled "The Cheerios Effect". State all the assumptions made, and any simplifications used or implied. Is there a characteristic separation distance beyond which no attraction or repulsion between plates is felt? Is there a dimensionless group that can be computed to estimate how strong or weak the capillary interaction forces can get for a given fluid? Will the attraction get stronger or weaker as the temperature of the fluid and the two plates are raised, such that no temperature gradients are created? (Advanced problem, for self-study only: Check if you can rederive the expressions outlined in this paper for attraction between spherical particles confined to the liquid/air interface).

(5) **Show the derivation for the size of a puddle** formed by a large drop on a solid substrate and of a **floating lens formed on an immiscible liquid substrate,** assuming the capillary length is exceeded in both cases**.**

(6) **"Can soft solids be shaped by capillary forces, just like ordinary liquids?"** (Required for graduate students, optional for undergraduates). Find and read Physical Review Letter 108, 094301, 2012. Write a critical review (one to two pages, Times New Roman, font size 12, single spacing) of the published paper without using AI help. (HINT: Summarize the key new ideas presented in this paper, check if the derivations are accurate, comment on any overly simplistic or unphysical assumptions made, and check their limits of applicability, list any papers the authors might have missed, and find any grammatical or mathematical errors herein. (Also see Soft Matter, 2012, 8, 7177, contact angle on soft surfaces, and related papers).

Provide only simple estimates/ expressions in problems 2-4. Comments on missing details, or how a more accurate analysis can be carried out, are welcome. In problem 5, you can write a longer review if you have something exceptional to contribute. Among other things, heroic effort is up for grabs for exceptional effort.

1 i.) The phenomena discussed in the video that was two types of phenomena described thru video that are going to be replicated at home to study and evaluate the phenomena are:

1. Surface Tension in Flowing Liquids

For this specific aspect of surface tension, the phenomena studied here is surface tension phenomena in flowing liquids. The study will show how changes in surface affect a flowing stream of water and how the change in the surface that has contact with the water will cause it to alter the flow of water.

Experiment: For the experiment describing surface tension in flowing water, a running stream of tap water that is moving smoothly replicating a thin film. The surface interacting with the water will be a finger. Two trials will take place where the finger is moved to the water. The trials will be done in a time interval of 10 seconds to observe the alterations of flow in water when interacting with the surface.

Results and connection to phenomena: After conducting the experiment, the results are that when the non-soap covered finger was added to the stream of water, the water would surround the finger and continue its stream as it left contact with the finger. Meanwhile when the finger covered in soap would cause the water stream to move rapidly away from the finger and disrupt the flow of the stream, making it spray away. The results of the experiment are conclusive of the phenomena of surface tension on running water from the video. When there is high surface tension, the non-soap finger trial flow will just surround the surface that the water is interacting with. Meanwhile, when surface tension is much lower, the surface tension causes the water to move irradicably from the stream. This is due to low surface tension caused by the soap being hydrophobic, repelling the water away, forcing the water to move away from the stream.

2.Surface Tension Gradients and Fluid Motion.

For the phenomena of tension gradients and fluid motion, the experiment will focus on how the between two liquids is altered when tension between surfaces is altered. In this case, the study of fats and water and hydrophilic substances will demonstrate how tension is affected by different substances and their reaction to changes in tension.

Experiment: In this experiment, a thin plate will fold a small amount of milk that will be the area of the experiment. In this case, fat is one of the liquids to demonstrate surface change in surface tension. Next, 4 drops of food coloring are added in different spots of milk. The food coloring will stay in the area hardly mixing with milk or other drops of food dye. Next, coat the soap to a cotton swab and add it to the milk with food coloring. Observe the changes in food coloring.

Results and connection to phenomena: After adding the cotton swab with soap, the surface tension changes since the soap lowers it. Once the soap was added, the food coloring would move away from the soap and bind with the rest of the food coloring. This is due to the hydrophobic nature of soap causing leading to the low surface tension leading to food coloring searching for an area with more surface tension. This made the milk or the fat create motion on the food coloring. The colors mixed in swirl and made a brown dark like color. This connects to how video would explain that changes in surface tension in fat would cause motion and move around. In this case, the surface tension change caused by the soap made the fat from the milk move around, missing the water it cannot mix with bind to gather as a single amount.

1 ii.) My favorite dimensionless number is the Reynolds Number (Re). The reason why it is my favorite number is due to how it is the basis for all transport phenomena calculations. Once the number is established, it can explore all aspects of design since it allows for the creation of all governing equations to explain fluid mechanics. The Reynolds number is found by using this equation. Once found, the number can describe the flow if it is laminar or turbulent. This later is used to find pressure drops and friction flow for designing plants.

1 iii). List of all characteristic lengths

All highlighted lengths are related to surface tension.

Λ	2D density	Defined per unit surface area	T^0/L^-2
Φ	1D density / normal stress coefficient	Acts along a line on a surface (contact lines)	T^0/L^-1

Σ	Stress; surface tension gradient	Directly represents surface tension and its spatial variation	T^-2/L^-1
Γ	Surface tension / stiffness / moduli	Measures interfacial stiffness and elasticity	T^-2/L^0
ζ	Surface viscosity; mobility	Governs flow along a surface	T^-1/L^0
η	Viscosity (surface contribution included)	Controls surface and interfacial shear	T^-1/L^-1
φ	Mass flux (including surface flux)	Transport along an interface	T^0/L^-1
Ψ	Force density / pressure gradient	Produces forces normal and tangential to surfaces	T^-2 /L^-2

Q2.) Butterfly Proboscis.

Find capillary rise w/ Jurin's Height eq.

$$R = H$$

$$H = \frac{2\gamma \cos\theta_E}{\rho g R} \Rightarrow R = \frac{2\gamma\cos\theta_E}{\rho g H}$$

assumptions needed $H \approx R$

$\theta_E = 0 = \cos\theta_E = \underline{1}$

$\gamma = 0.72\ N/m$, @ RT

$\gamma = 0.72\ mN/m$

$\rho = 1000\ kg/m^3$

$$R = \frac{2(0.72\ N/m)}{1000\ \frac{kg}{m^3} \cdot 9.81\ m/s^2 \cdot H}$$

$$\rightarrow R = \frac{0.144\ N/m}{9800\ kg/m^2 \cdot s^2 \cdot H} \Rightarrow$$

$H = 1\ mm = 0.001\ m$

$$R = \frac{0.144\ N/m}{9800\ kg/m^2 \cdot s\ (0.001 m)} = \boxed{0.0147 m \text{ or } 1.47 cm}$$

$H = 10\ mm = 0.01\ m$

$$R = \frac{0.144\ N/m}{9800\ kg/m^2 \cdot s\ (0.01 m)} = \boxed{0.0147 m \text{ or } 1.47 mm}$$

$h = 100mm = 0.1\ m$

$$R = \frac{0.144\ N/m}{4800\ kg/m^2 \cdot s\ (0.1m)} = \boxed{0.000147m \text{ or } 0.147mm}$$

Calc for Different Height.

$P = \rho g h$

$H = 0.1mm = 0.0001m$

$P = (1000 \frac{kg}{m^2})(9.81 \frac{m}{s^2})(0.0001m)$

$P = 0.981\ Pa$

$H = 5mm = 0.005m$

$P = (1000 \frac{kg}{m^2})(9.81 \frac{m}{s^2})(0.005m)$

$P = 49.05\ Pa$

$H = 50mm = 0.05m$

$P = (1000 \frac{kg}{m^2})(9.81 \frac{m}{s^2})(0.05\ m)$

$P = 490.5\ Pa$

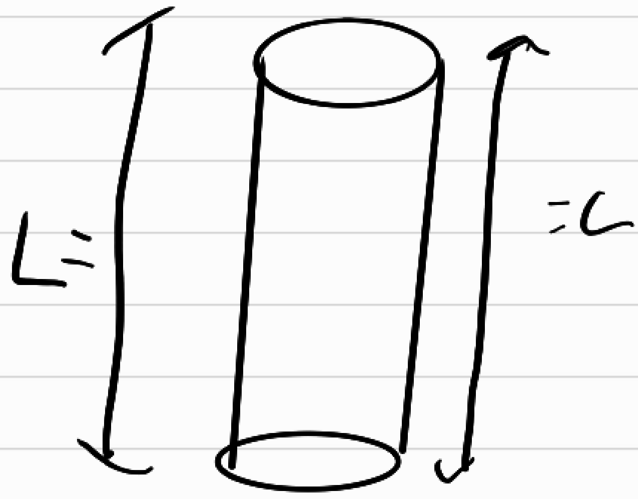
Comparing to Atm Pressure

h	Hydrostatic Pressure	% of Atmo Pres
0.1mm	0.981 Pa	$9.68 \cdot 10^{-4}$ %
5 mm	49.05 Pa	0.048%
50 mm	490.5 Pa	0.48%

Q3). water Striders

Surface tension:

$F_\gamma = \gamma \cdot$ (contact surface area)



Upward force $= F_\gamma = 2\gamma L$

max weight supports

$W_{max} = 2\gamma L$

W_{max} at Unit Length

$$\frac{W_{max}}{L} = 2\gamma$$

Using Room temp γ, ($\gamma = 0.072\ N/m$)

$$\frac{W_{max}}{L} = 2\gamma = 2(0.072\ N/m)$$

$= 0.144\ N/m$

proves as $W_{max} = 2\gamma L$

b). Critical Buckling Length

Under Capillary Comp

$F = 2\gamma L$

$$F_{cr} = \frac{\pi^2 E \cdot I}{L^2}$$

When : $I = \frac{\pi R^4}{4}$

$$2\gamma L = \frac{\pi^2 E I}{L^2}$$

apply L^2 to Both

$$2\gamma L^3 = \pi E I$$

apply I $2\gamma L^3 = \pi^2 E \left(\frac{\pi R^4}{4} \right.$

$$L^3 = \frac{\pi^3 E R^4}{8\gamma}$$

$$L_{cr} = \left(\frac{\pi^3 E R^4}{8\gamma} \right)^{1/3}$$

Human Hair values:

$E = 4 GPa = 4 \cdot 10^9 Pa$

$R = 20 \mu m = 20 \cdot 10^{-6} m$

$\gamma = 0.072 N/m$

$$I = \frac{\pi R^4}{4}$$

$R^4 = (20 \cdot 10^{-6})^4 = 1.6 \cdot 10^{-19}$

$I = 1.256 \cdot 10^{-19} m^4$

$$L^3 = \frac{\pi^2 EI}{2\gamma}$$

$$L^3 = \frac{\pi^2 (4 \cdot 10^9) \cdot (1.256 \cdot 10^{-19} m^4)}{2 \cdot 0.072 N/m}$$

$$L^{3/3} = (3.44 \cdot 10^{-3})^{1/3}$$

$$L_{cr} = 3.3 \text{ mm}$$

compare to water

$\gamma = 0.072$ N/m
$\rho = 1000$ kg/m^3
$g = 9.81$ m/s

$$l_c = \sqrt{\frac{\gamma}{\rho g}} = \sqrt{\frac{0.072 N/m}{1000 kg/m^3 \cdot 9.81 m/s^2}}$$

when l_c is < 3 mm

Surface tension is stronger Than gravity, leads to Floating like water striders and human hair.

when $l_c > 3$ mm. Gravity Beats Surface tension makes it sink

Q4) "Cheerro Effect"

$dy/dx \ll 1$

$$\gamma \frac{d^2h}{dx^2} = \rho g h$$

rewrite w/ least as dh/dx

$$\frac{d^2h}{dx^2} = \frac{h}{L_c^2}$$

$$L_c = \sqrt{\frac{\gamma}{\rho g}}$$

apply to Regions

$$h(x) = A e^{-x/L_c} + B e^{x/L_c}$$

left: $A = -L_c \cot\theta_1 e^{x/L_c} = h_1$

right $= B = -L_c \cot\theta_2 e^{(d-x)/L_c} = h_3$

apply Between plates

h_2 $$h(x) = \frac{L_c}{\sinh d/L_c}\left[\cot\theta_1 \cosh\left(\frac{d-x}{L_c}\right) + \cot\theta_2 \cosh\left(\frac{x}{L_c}\right)\right]$$

Applying to horizontal forces.

$P = \rho g h$

left h_2) $$F_h = \frac{1}{2}\rho g\left[h_1(0)^2 - h_2(0)^2\right]$$

add height

$$F_h = \frac{\gamma}{2}\frac{\left(\cot\theta_1 \cosh\frac{d}{L_c} - \cot\theta_2\right)^2}{\sinh^2(d/L_c)} - \frac{\gamma}{2}\cot\theta_1$$

Applying The Cheerio effect

$$\underset{W}{\text{Weight}} = \underset{ST}{\text{Surface tension}} + \underset{B}{\text{Buoyancy}}$$

$$\frac{4}{3}\pi \rho_b g R^3 - 2\pi \gamma R \sin\psi_c \frac{z_c'}{\sqrt{1+z_c'^2}} + B_0$$

Apply small slopes + Bond num

Bond num + Slope

$$B = \frac{\rho g R^2}{\gamma} \ll 1$$

Combine w/ prev Eq

$$z_c' \sin\psi_c = B\left(2D - 1 + \frac{1}{2}\cos\theta - \frac{1}{6}\cos^3\theta\right)$$

resultant weight W_{ep}

$$W_{ep} = 2\pi\gamma R B$$

Axis-symetric Equation.

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dh}{dr}\right) - \frac{h}{L_c^2}$$

Apply w prev Eq:

$$h(r) = -z_c' \sin\psi_c R K_0 \, r/L_c$$

K_0 = Bessel function from paper.

Interaction Energy:

Use Nicolson Superposition

$$E(l) = -(W_{ep})h(l)$$

$$E(l) = -2\pi \gamma R^2 B^2 K_0\left(\frac{l}{L_c}\right)$$

Interaction Force

$$F(l) = -\frac{dE}{dl}$$

$$\frac{d}{dx} K_0(x) = -K_1(x)$$

$$F(l) = -2\pi \gamma R B^{\frac{5}{2}} \cdot K_1\left(\frac{l}{L_c}\right)$$

Characteristics of Separation Distance

$l \gg L_c$ as Bessel asymptotic

$$K_1(x) = \sqrt{\frac{\pi}{2x}}\, e^{-x}$$

simplify to

$$F(l) = e^{-l/L_c}$$

From prev problem text

L_c as Interaction as Boundary
For floating w surface tension or
Sinking due to gravity 2.7mm

Effect on Particle size;

$F \propto B^2 \propto R^4$ .. This

w theory from paper: Larger radius
= more force, therefore for this
the following must happen:

1. Nicolson Approx fails
2. non Linear Laplace
3. Pressure Capillary Solution is

Stronger Then Buoyancy.

From paper

$$F < B^2 < B$$

$$100 < B^2 < 10000$$

source

$$10 < B < 100$$

Applying to temp effect.

using $L_c = \left(\frac{\gamma}{\rho g}\right)$ as given eq

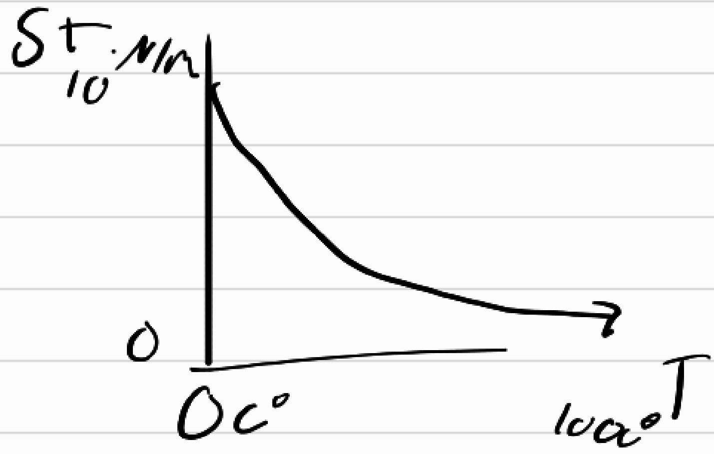
that states all higher temp causes Bonds to weaken. Therefore lower Surface Tension.

Using temp relation to Surface tension to make a gradient.

$$\gamma = \gamma(T(x))$$

$$\nabla_s \gamma$$

Heat is related to Surface tension, tension is lower as Heat rises



Q 5.) Puddle derivation

Puddle size is related to surface Tension from capillary length of a solid & liquid interaction.

$R \gg \kappa^{-1}$ using theory

1. Surface Forces $\gamma_{so} - (\gamma + \gamma_{SL})$
2. Hydrostatic pressure $\bar{p}$ is integrated over entire thickness of the liquid.

$$\bar{p} = \int_0^e \rho g (e - \bar{z})\, d\bar{z} = \frac{1}{2}\rho g e^2$$

applying Equilibrium

$$\frac{1}{2}\rho g e^2 + \gamma_{so} - (\gamma + \gamma_{SL}) = 0$$

rewrite as

$$S = \frac{1}{2}\rho g e^2$$

Use Young's Law to describe the equilibrium of forces acting on the line of contact

$$\gamma_{sc} - (\gamma \cos\theta_2 + \gamma_{SL}) = 0$$

$$\therefore \gamma \cdot (1 - \cos\theta_E) = \frac{1}{2}\rho g e^2$$

$$2\frac{\gamma}{\rho g}(1 - \cos\theta_E) = e^2$$

$\sin\theta_E$

rewrite as

$$2\sqrt{\frac{\gamma}{\rho g}}\ \sin\frac{\theta_E}{2} = e$$

or as $\boxed{2\kappa^{-1}\sin\left(\frac{\theta_E}{2}\right) = e}$

applying that a puddle is a large drop on a surface

using $(R \gg \kappa^{-1}_{AB}, \kappa_A^{-1})$

$$F(e) = \left[\frac{1}{2}\rho_A g(e-e')^2 - \frac{1}{2}(\rho_B - \rho_A)g e'^2 - S\right]\frac{\Omega}{e}$$

apply Archimeds condition

$$\rho_A g e = \rho_B g e'$$

$$e' = \frac{\rho_A}{\rho_B} e$$

Define $\bar{\rho} = \frac{\rho_A}{\rho_B} \cdot \rho_B - \rho_A$

$$F_e = \left[\frac{1}{2}\bar{\rho} g e^2 - S\right]\frac{\Omega}{e}$$

Differentiate to get to pure puddle.

$$F(e) = \Omega\left[\frac{1}{2}\bar{\rho} g e - \frac{S}{e}\right]$$

$$\frac{dF}{de} = \Omega\left[\frac{1}{2}\rho g + \frac{S}{e^2}\right]$$

$$0 = \Omega\left[\frac{1}{2}\bar{\rho} g + \frac{S}{e^2}\right]$$

$$0 = \frac{1}{2}\rho g + \frac{S}{e^2}$$

$$-\frac{S}{e^2} = \frac{1}{2}\rho g$$

$$-S = \frac{1}{2}\rho g e^2$$

$$\boxed{e = \sqrt{\frac{2 \cdot S}{\rho g}}}$$